

April 27, 2026

Recall, if $\vec{x}: [a, b] \rightarrow \mathbb{R}^n$ is a piecewise diff. path and $\vec{y}: [c, d] \rightarrow \mathbb{R}^n$ is a piecewise diff. path then $\vec{y}$ is a reparametrization of $\vec{x}$ if there is a continuous one-to-one and onto function $g: [c, d] \rightarrow [a, b]$ such that $\vec{y}(t) = \vec{x}(g(t))$.

Theorem: Let $\vec{x}: [a, b] \rightarrow \mathbb{R}^n$ be a piecewise differentiable path and $f: \mathcal{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ a cont. function such that $\vec{x}(t) \in \mathcal{X}$ for all $t \in [a, b]$.

If $\vec{y}: [c, d] \rightarrow \mathbb{R}^n$ is any reparametrization of $\vec{x}$, then

$$\int_{\vec{y}} f \, ds = \int_{\vec{x}} f \, ds.$$

def: If $C \subseteq \mathbb{R}^n$ is a curve that can be parametrized by a piecewise differentiable path, and $f: \mathcal{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuous function such that $C \subseteq \mathcal{X}$, then we define the integral of f over the curve C by

$$\int_C f \, ds = \int_{\vec{x}} f \, ds$$

(where $\vec{x}$ is any piecewise differentiable parametrization of C).

e.g. $f(x, y) = x$ over the arc C on the unit circle between angles π and 2π

$$\vec{x}(t) = (t, -\sqrt{1-t^2}) \quad -1 \leq t \leq 1$$

$$\|\vec{x}(t)\| = \sqrt{t^2 + 1-t^2} = 1$$

$$f(\vec{x}(t)) = t$$

$$\int_C f \, ds = \int_{-1}^1 t(1) \, dt = \left. \frac{t^2}{2} \right|_{-1}^1 = 0$$

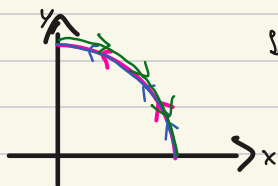
Recall, if $\vec{x}: [a, b] \rightarrow \mathbb{R}^n$ is a piecewise diff. path and $\vec{y}: [c, d] \rightarrow \mathbb{R}^n$ is a piecewise diff. path then $\vec{y}$ is a reparametrization of $\vec{x}$ if there is a continuous one-to-one and onto function $g: [c, d] \rightarrow [a, b]$ such that $\vec{y}(t) = \vec{x}(g(t))$.

Proposition: If $f: [c, d] \rightarrow \mathbb{R}$ is a continuous and one-to-one function, then either f is strictly increasing or f is strictly decreasing.

def: Let $\vec{x}: [a, b] \rightarrow \mathbb{R}^n$ be a path and $\vec{y}: [c, d] \rightarrow \mathbb{R}^n$ be a reparametrization of $\vec{x}$ (with $\vec{y}(t) = \vec{x}(g(t))$), then we say:

- 1) $\vec{y}$ orientation preserving if $\vec{y}(c) = \vec{x}(a)$ (so g is increasing)
- 2) $\vec{y}$ orientation reversing if $\vec{y}(c) = \vec{x}(b)$ (so g is decreasing)

e.g. $\vec{x}(t) = (\cos t, \sin t) \quad 0 \leq t \leq \pi/2$
 and $\vec{y}(t) = (\cos(2t), \sin(2t)) \quad 0 \leq t \leq \pi/4$
 and $\vec{z}(t) = (\sin t, \cos t) \quad 0 \leq t \leq \pi/2$



Let's find $f: [0, \pi/4] \rightarrow [0, \pi/2]$
 such that $\vec{y}(t) = \vec{x}(f(t))$
 $f(t) = 2t$
 $\vec{y}(t) = (\cos(2t), \sin(2t)) = \vec{x}(2t)$
 this shows $\vec{y}$ is a reparam. of $\vec{x}$

Let's find $g: [0, \pi/2] \rightarrow [0, \pi/2]$ such that $\vec{z}(t) = \vec{x}(g(t))$.
 $g(t) = \pi/2 - t$

$$\begin{aligned} \vec{z}(t) &= (\sin t, \cos t) = (-\sin(-t), \cos(-t)) \\ &= (-\cos(-t - \pi/2), \sin(\pi/2 - t)) \\ &= (\cos(\pi - t - \pi/2), \sin(\pi/2 - t)) \\ &= \vec{x}(g(t)) \end{aligned}$$

$$\begin{aligned} * \sin t &= -\sin(-t) \\ \sin(t + \pi/2) &= \cos t \\ \sin t &= \cos(t - \pi/2) \\ \cos(t + \pi) &= -\cos t \end{aligned}$$

Theorem: Let $\vec{x}: [a, b] \rightarrow \mathbb{R}^n$ be a piecewise differentiable path and $\vec{F}: \mathcal{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ a cont. vec. field such that $\vec{x}(t) \in \mathcal{X}$ for all $t \in [a, b]$.

If $\vec{\gamma}: [c, d] \rightarrow \mathbb{R}^n$ is any reparametrization of $\vec{x}$, then either

1) If $\vec{\gamma}$ is orientation preserving, then $\int_{\vec{x}} \vec{F} \cdot d\vec{s} = \int_{\vec{\gamma}} \vec{F} \cdot d\vec{s}$

2) If $\vec{\gamma}$ is orientation reversing, then $\int_{\vec{x}} \vec{F} \cdot d\vec{s} = - \int_{\vec{\gamma}} \vec{F} \cdot d\vec{s}$.

def: If $C \subseteq \mathbb{R}^n$ is an oriented curve which can be parametrized by a piecewise diff. path., then we define the vector line integral over C

$$\int_C \vec{F} \cdot d\vec{s} = \int_{\vec{x}} \vec{F} \cdot d\vec{s} \quad \text{for any orientation preserving parametrization } \vec{x} \text{ of } C.$$

e.g. Let C be the line segment from $(1, 2)$ to $(3, 6)$ and compute

$$\int_C y dx + (x+y) dy$$

$$dx = \dot{x} dt \quad dy = \dot{y} dt$$

$$\vec{x}(t) = (1, 2) + t [(3, 6) - (1, 2)] = (1+2t, 2+4t) \quad 0 \leq t \leq 1$$

$$\vec{x}'(t) = (2, 4)$$

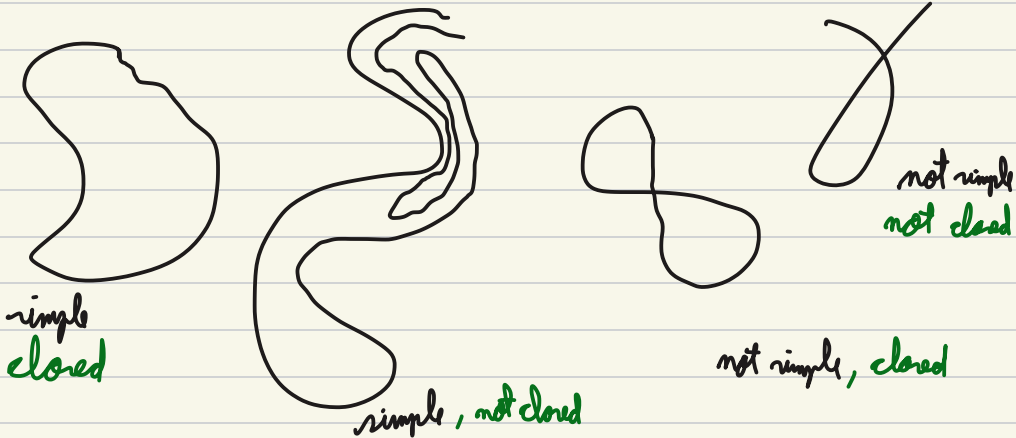
$$\begin{aligned} \int_C y dx + (x+y) dy &= \int_0^1 (2+4t) 2 dt + \int_0^1 (3+6t) 4 dt \\ &= \int_0^1 [4 + 12 + 8t + 24t] dt = \int_0^1 (16 + 32t) dt = 16t + 16t^2 \Big|_{t=0}^{t=1} \\ &= 16 + 16 \end{aligned}$$

Simple and closed curves

def: 1) A cont. path $\vec{x}: [a, b] \rightarrow \mathbb{R}^n$ is called simple if $\vec{x}$ has no self-intersections except possibly $\vec{x}(a) = \vec{x}(b)$

2) A curve $C \subseteq \mathbb{R}^n$ is called simple if it is the image of a simple path

e. g.

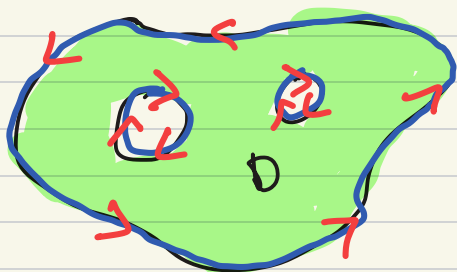


def: 1) A cont. path $\vec{x}: [a, b] \rightarrow \mathbb{R}^n$ is called closed if $\vec{x}(a) = \vec{x}(b)$.

2) A curve $C \subseteq \mathbb{R}^n$ is called closed if it is the image of a closed path

def: Let $D \subseteq \mathbb{R}^2$ be a compact region (D is closed and bounded), we define the boundary of D as
 $\partial D = \{ \vec{x} \in D : \vec{x} \text{ is not an interior point} \}$

e.g.



• ∂D

Theorem (Green's Theorem): Suppose D is a compact region in $\mathbb{R}^2$ with boundary $C = \partial D$ consisting of finitely many simple closed piecewise diff. curves.

Give C an orientation such that if we travel C following the orientation, the interior of the region is always to the left.

Suppose $F(x, y) = (P(x, y), Q(x, y))$ is a C^1 vector field, then

$$\oint_C P(x, y) dx + Q(x, y) dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

